



BASIC DRAFTING



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GOALS OF TECHNICAL DRAWING



- Detail drawing must be accurate.
- Detail drawing must be executed with proper technique or good workmanship.
- Detail drawing should be neat.
- Detail drawing must be made with speed.

TWO CLASSES OF DRAWING

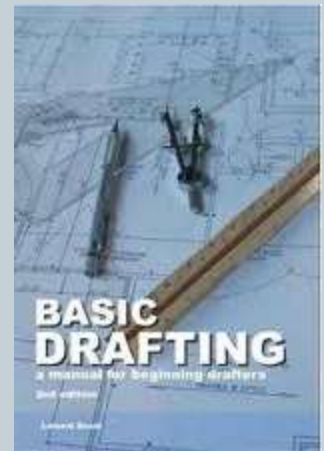


- **ARTISTIC-** The artist expresses aesthetic ideas or emotions.
- **TECHNICAL-** Exact means of expression and accuracy is one of the main objective.

DEFINITION AND RELATIONSHIP OF;



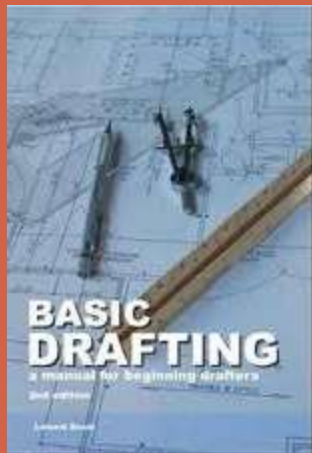
- PAINTING
- DRAWING
- DRAFTING
- SKETCHING





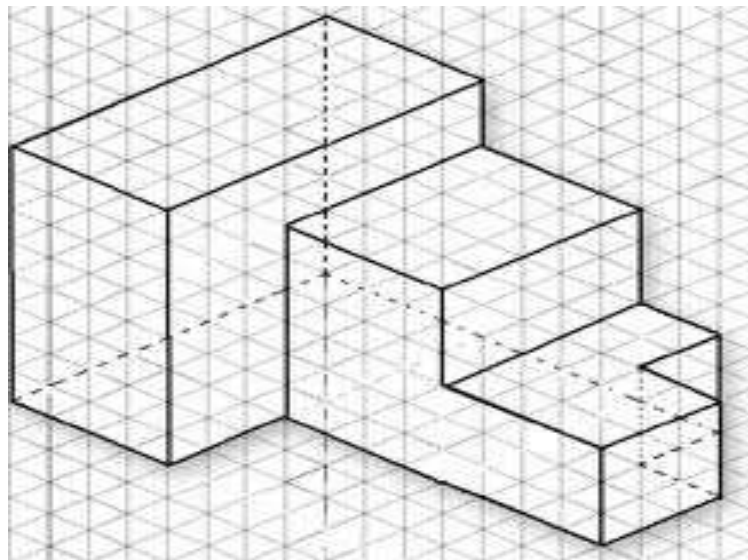
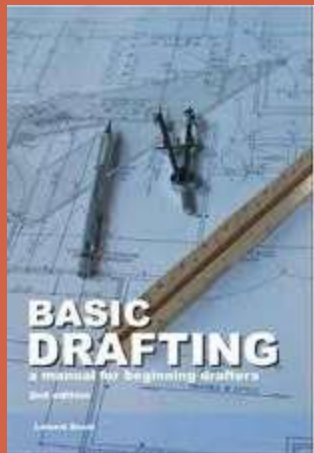
PAINTING

Is a form of visual art that uses wet media.



DRAWING

Is a form of visual art that uses dry media. It has two types: Technical and Artistic.

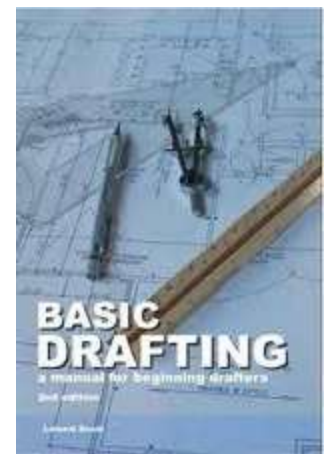




DRAFTING OR TECHNICAL DRAWING

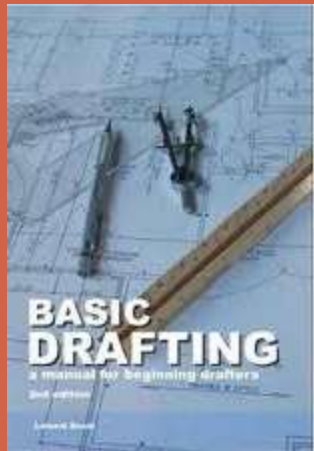
Is the act and discipline of composing plans that visually communicate how something functions or has to be constructed. Drafting is the visual language of industry and engineering. A ***drafter, draftsperson, or draftsman*** is a person who makes a technical drawing. A professional drafter who makes technical drawings is sometimes called a ***drafting technician***.

Drafting is the term used to refer to the systematic representation and dimensional specification of mechanical and architectural structures.



SKETCHING

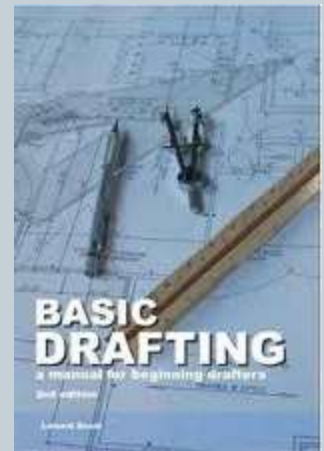
Is a quickly executed freehand drawing that is not intended as a finished work. In general, sketching is a quick way to record an idea for later use.



MATERIALS NEEDED FOR DRAWING



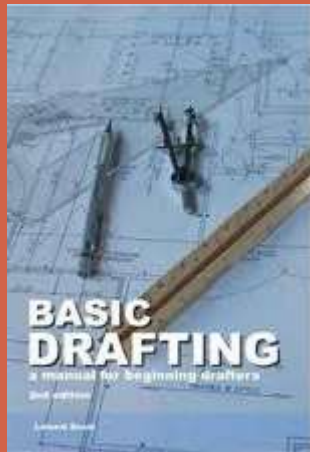
- DRAWING BOARD/TABLE
- DRAWING PAPER/PLATES
- TECHNICAL PEN
- MECHANICAL PENCIL
- ERASER
- COMPASS
- T-SQUARE
- TRIANGLES (45x45 AND 30x60)
- RULER
- PROTRACTOR





DRAWING BOARD

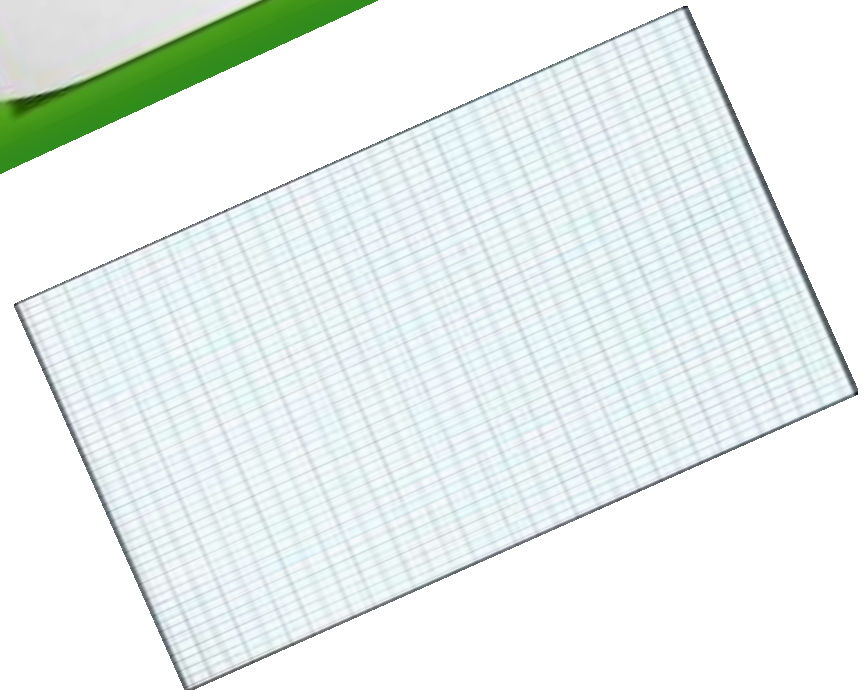
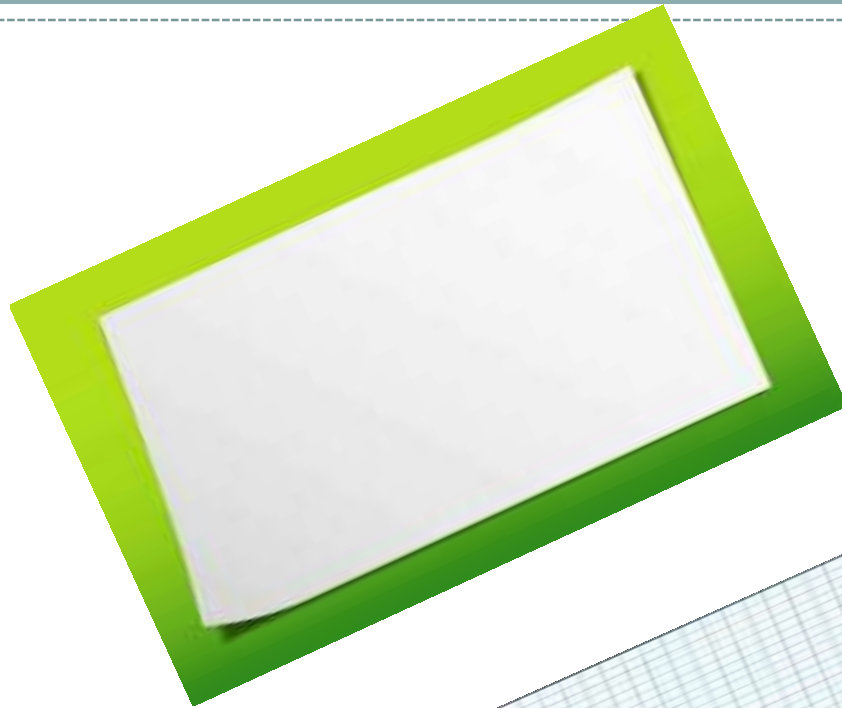
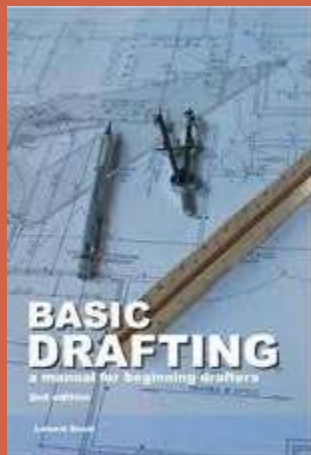
Base for drawing.





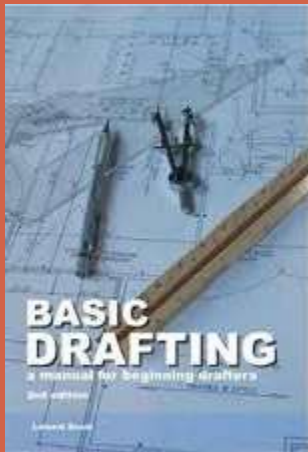
DRAWING PAPER

A drafting media for
showing the object.



TECHNICAL PEN

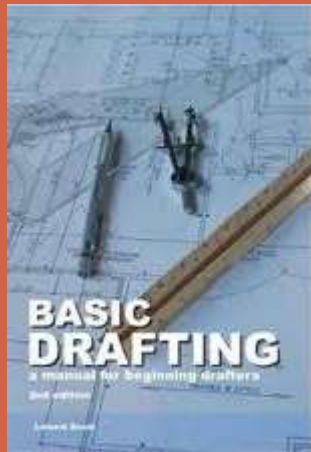
A drafting media for illustrating the object.





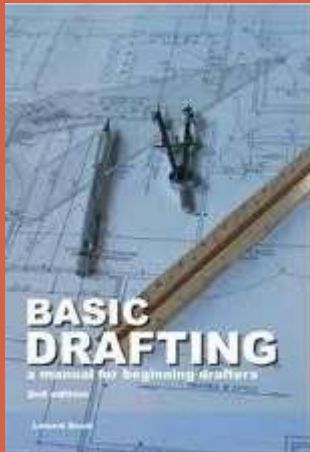
MECHANICAL PENCIL

A drafting media for freehand guides or sketching the object.



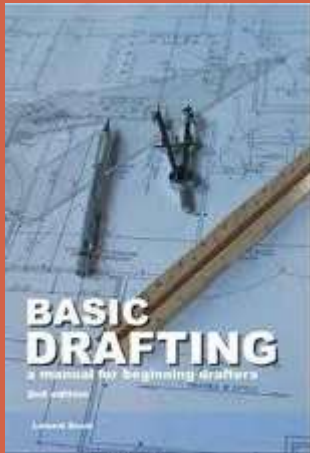
ERASER

Used to erase the unnecessary lines done by sketching.



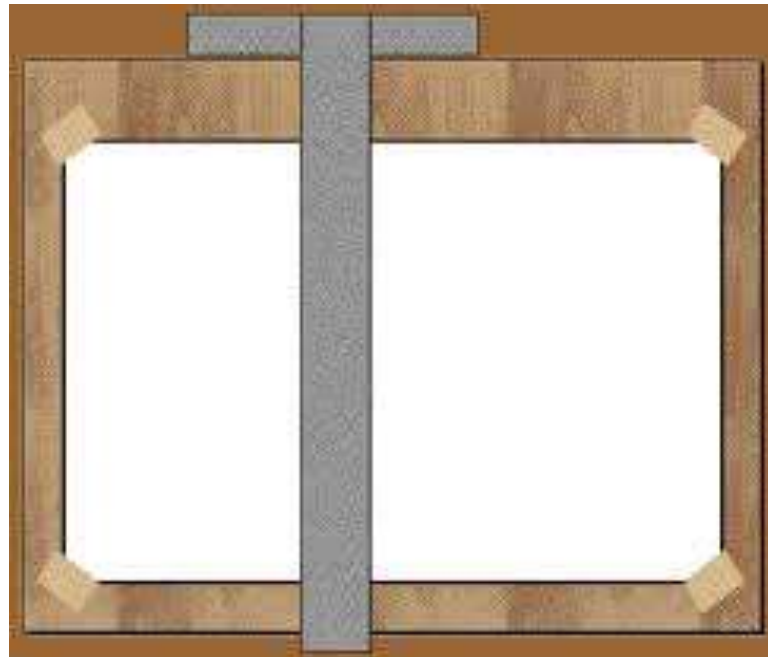
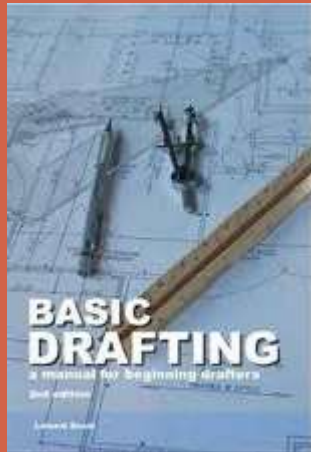
COMPASS

Used to draw arcs and circles.



T-SQUARE

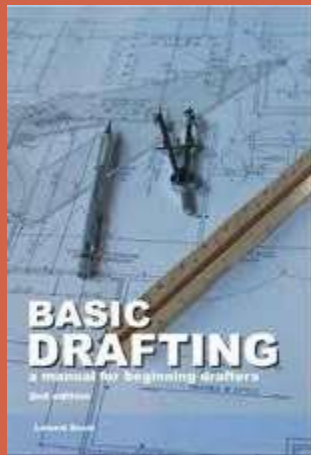
Used to draw horizontal and vertical lines and serves as base material for triangles.





TRIANGLES (30X60 AND 45X45)

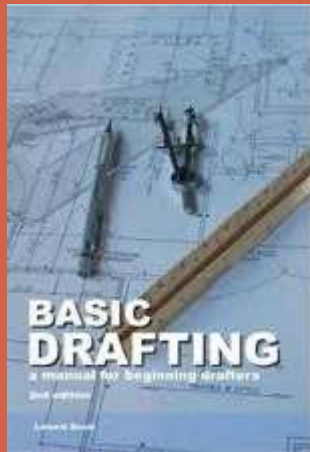
Used to draw inclined lines with proper angle and vertical lines.





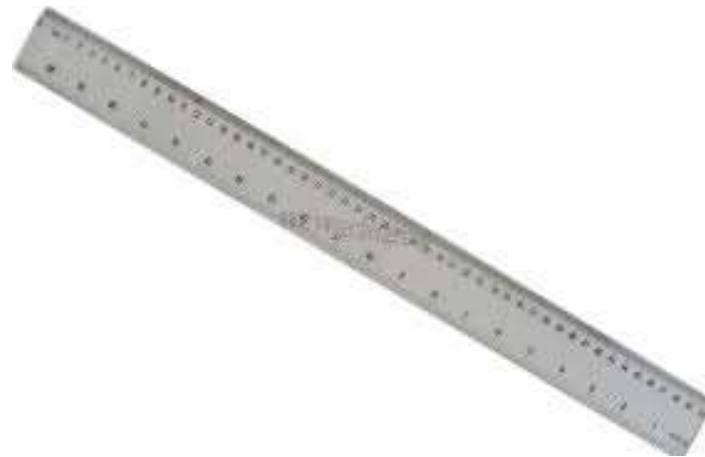
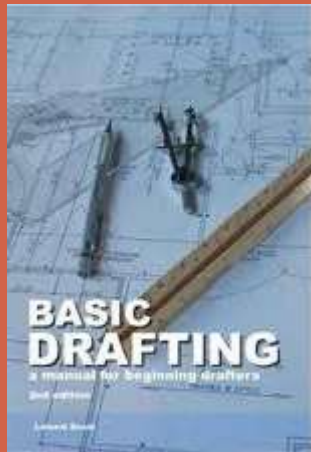
TRIANGULAR SCALE

Used to for measuring
an object or lines.



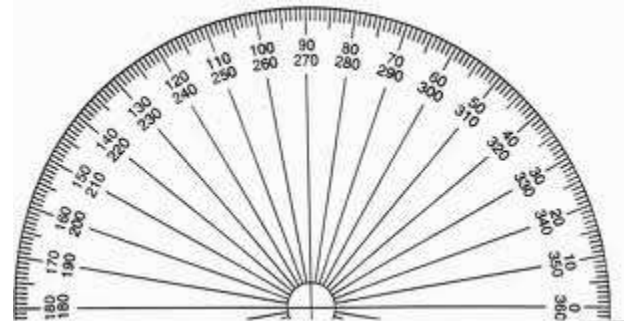
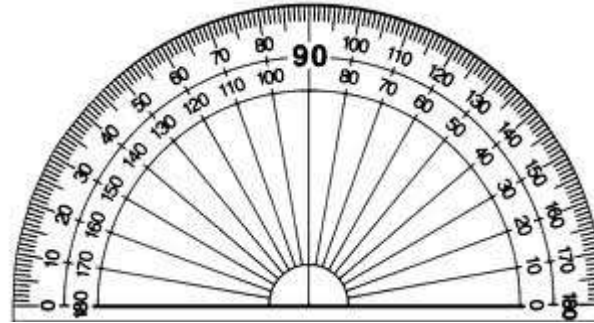
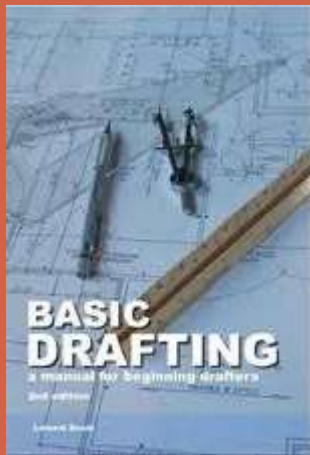
RULER

A straight edge material with measurement graduation used to make and measure dimensions.



PROTRACTOR

Used to measure angles.



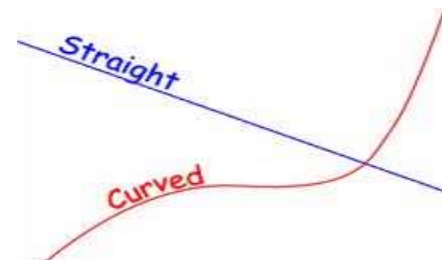
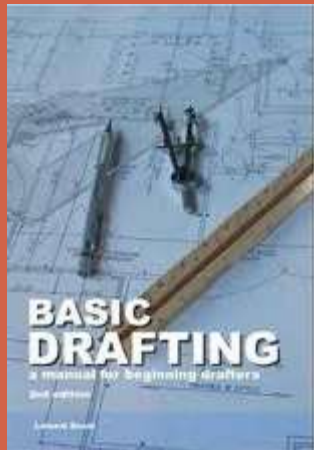
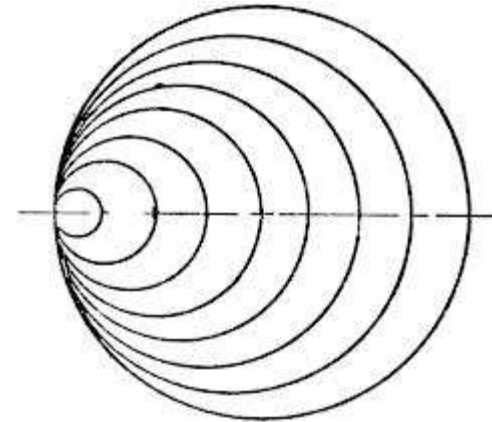
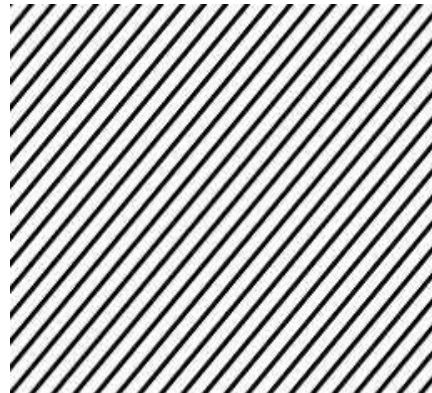
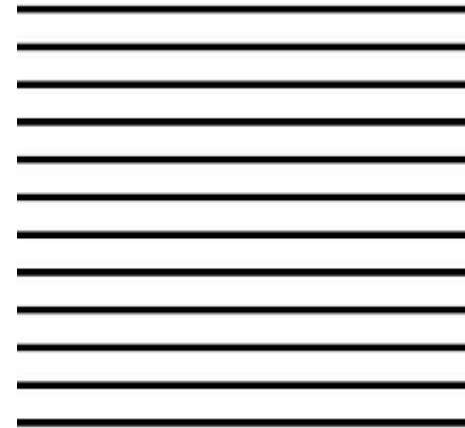
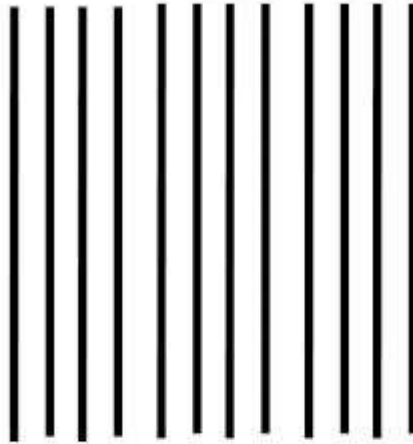
TYPES OF LINES

VERTICAL LINES

HORIZONTAL LINES

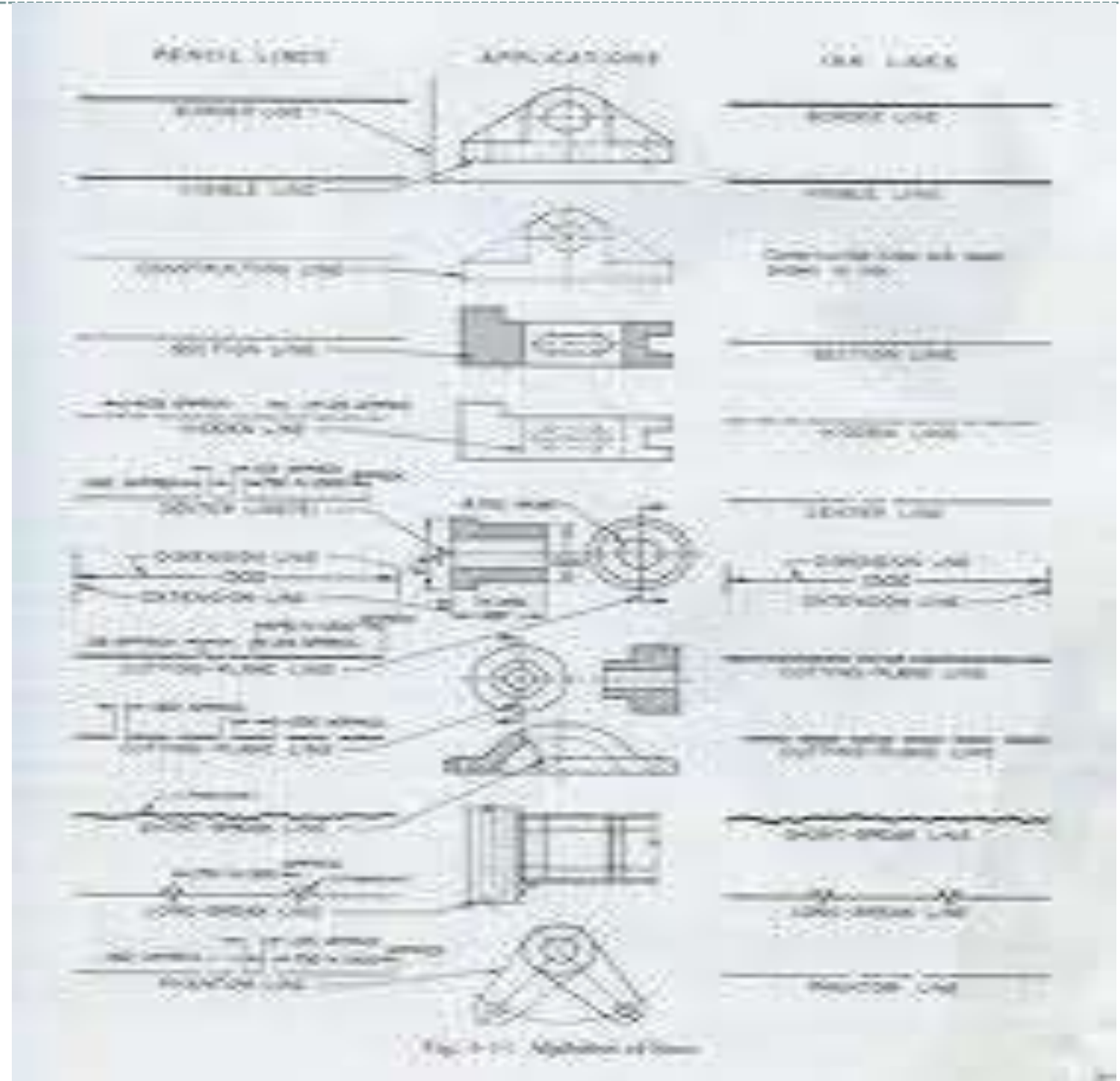
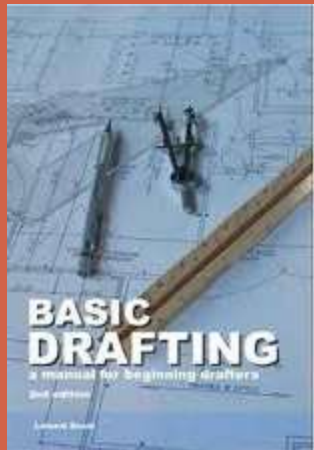
INCLINED LINES

CURVED LINES





Are lines used in drafting to identify the types of lines; thick, thin and medium. They have the same relation to drawings as letters do to words.



ALPHABET OF LINES

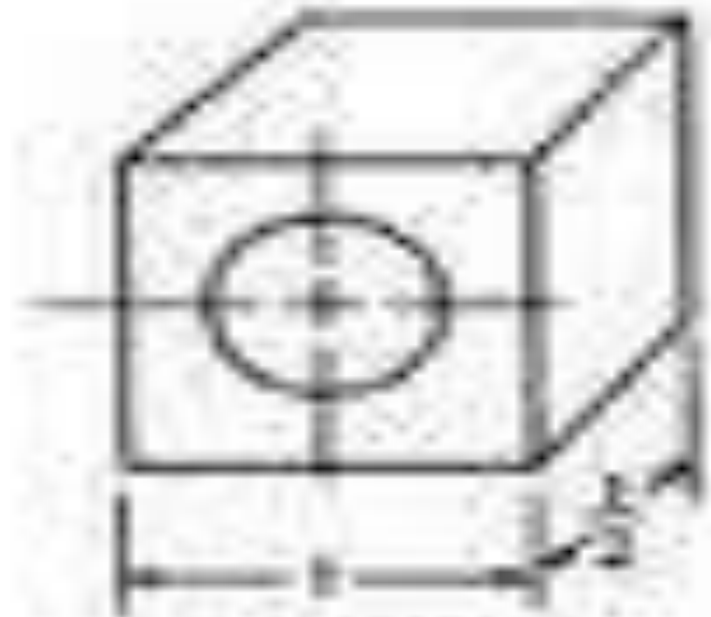


- **Visible or object line-** represents the visible edges of an object.
- **Hidden or invisible line-** represents the hidden edged of an object. It is made of short dashes.
- **Center line-** is the lightest line used to show the center of the object.
- **Dimension line-** used to indicate the measurement of objects.
- **Leader line-** A fine broken line with an arrowhead at one end used for indicating the measurement and name of a part.
- **Reference line-** An irregular fine curved line with an arrowhead at one end used to label the parts of an object.
- **Long break line-** A limiting line used to limit the length of an elongated object so that it can be shown without changing the size of its view.
- **Short break line-** Is similar in use with long break line. It is commonly used to limit the length of a cylindrical object.
- **Cutting plane line-** consist of long and short dashes used to indicate the cut view of objects.
- **Section line-** is an imaginary cut taken through an object so as to travel a shape or interior construction.

ALPHABET OF LINES



CAVALIER
PROJECTION



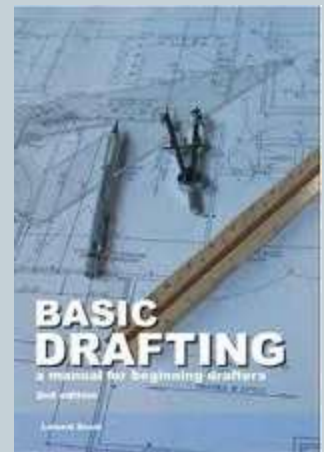
CABINET
PROJECTION

ISOMETRIC

LETTERING



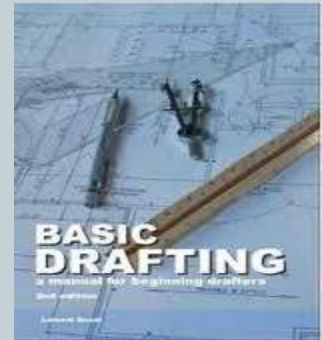
- Is a fundamental part of technical drawing.
- It is used for dimensions, notes and specifications to communicate with the fabricator.



BASIC CLASSIFICATION OF LETTERS



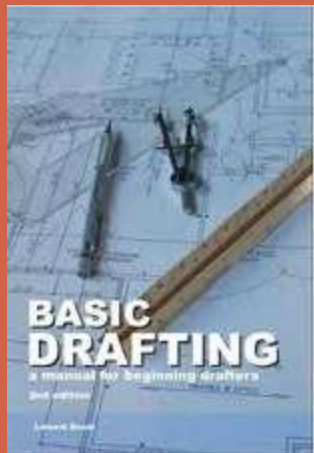
- ROMAN- all letters having strokes “accented” or made up of thick and thin elements.
- GOTHIC- all letters composed of uniform width elements.
- TEXT- Text lettering is one of the oldest letters styles in use. Text letters consist of solid main stems.
 - ITALIC- All slanting or inclined letters are classified as italic style.





ROMAN LETTERS

all letters having
strokes “accented” or
made up of thick and
thin elements.

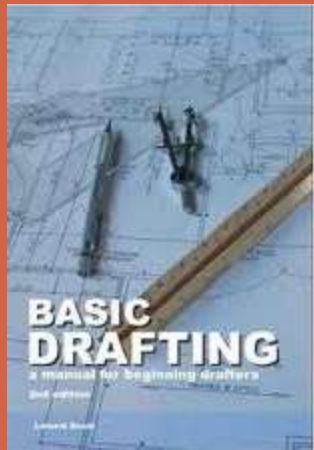


0123456789 —
ABCDEFGHIJKLMN
OPQRSTUVWXYZ
abcdefghijklmno
pqrstuvwxyz

GOTHIC LETTERS

all letters composed of
uniform width
elements.

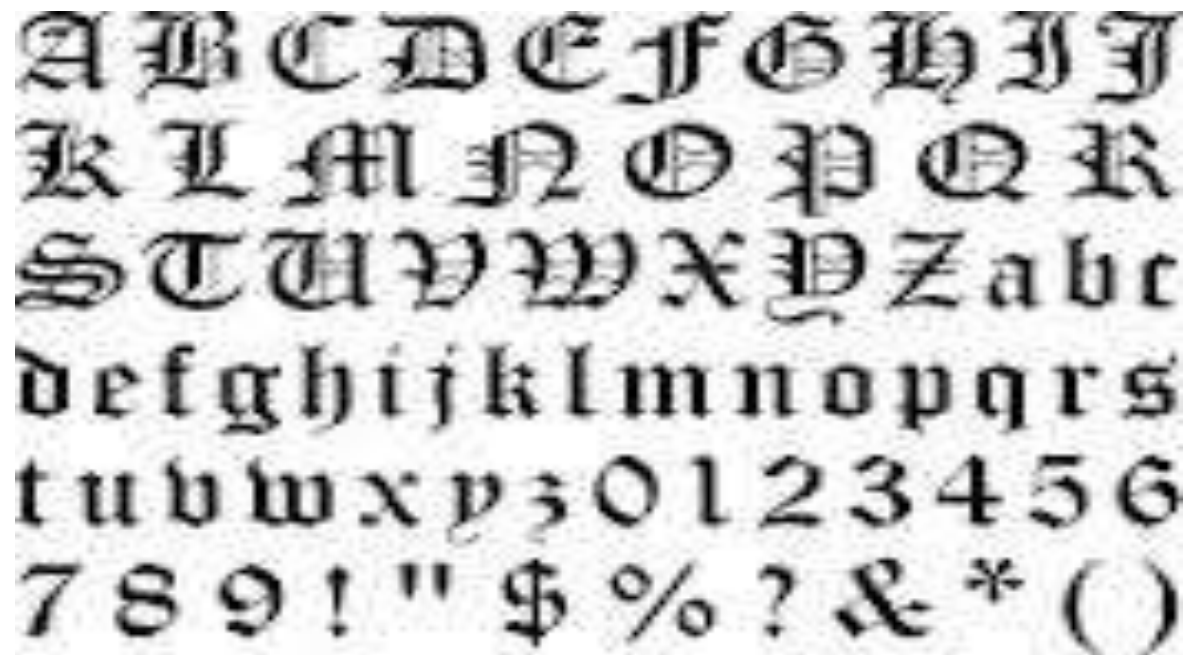
A B C D E F G H I J K L M N
O P Q R S T U V W X Y Z
1 2 3 4 5 6 7 8 9 0 & 2 $\frac{5}{8}$





TEXT LETTERS

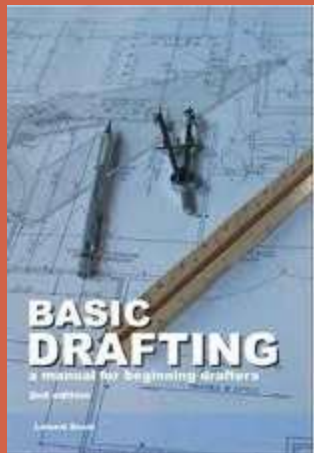
Text style letters are synonymous with Old English letters. Text lettering is one of the oldest letters styles in use. Text letters consist of solid main stems with solid block serifs. Text letters have strong vertical lines. Do not use text letters in all caps or words will be illegible. Limit the use of text letters to layouts that require religious or formal presentations.



ITALIC LETTERS

All slanting or inclined letters are classified as italic style.

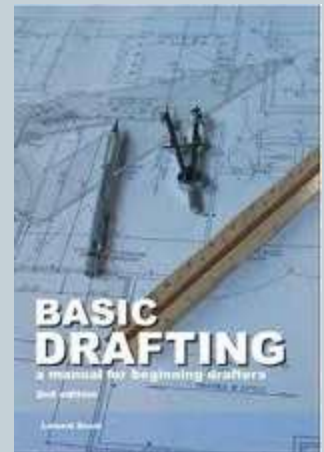
LETTERING



UNIFORMITY AND DESIGN IN LETTERING



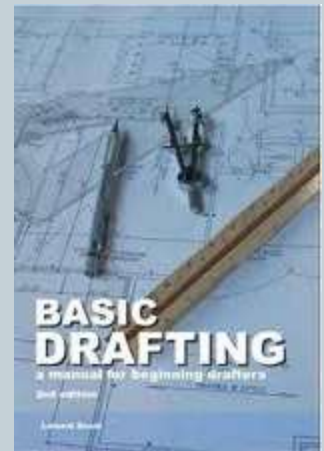
- In any style of lettering, uniformity is essential. Uniformity in height, proportion, inclination, strength of lines, spacing of letters and spacing of words ensures a pleasing appearance. Uniformity in height and inclination is promoted by the use of light guidelines



PROPORTIONS OF LETTERS



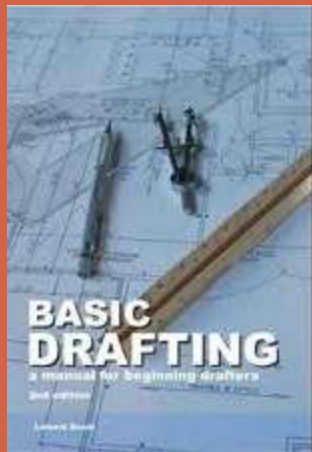
- NORMAL LETTERS
- EXTENDED LETTERS
- CONDENSED LETTERS
- LIGHTFACE
- BOLDFACE LETTERS



NORMAL LETTERS

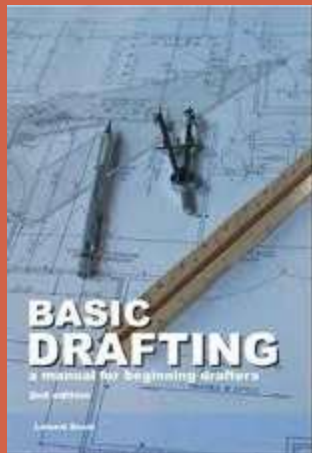
These are used when the space for lettering is adequate. They are neither too narrow nor too wide and have the same height.

A	B	C	D	E	F	G	H	I
J	K	L	M	N	O	P	Q	R
S	T	U	V	W	X	Y	Z	



EXTENDED LETTERS

Are letters wider than normal letters and are used when the space to be used is wider than normal.



WIDE LOAD
INTERSTATE BLACK

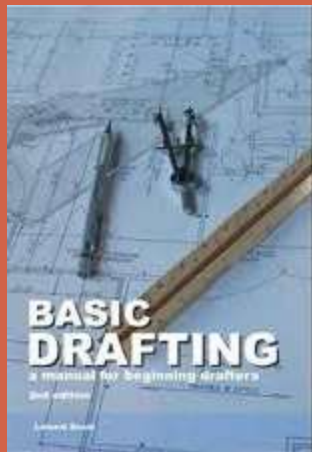
TIGHT WAD
INTERSTATE BLACK CONDENSED

**WIDE
LOAD**
TYPE CRIME: HORIZONTAL SCALE: 250%

TIGHT WAD
TYPE CRIME: VERTICAL SCALE: 200%

CONDENSED LETTERS

Letters that are narrower than normal. Used when space is limited.



WIDE LOAD

INTERSTATE BLACK

The set width is the body of the letter plus the space beside it.

TIGHT WAD

INTERSTATE BLACK COMPRESSED

The letters in the compressed version of the typeface have a narrower set width.

WIDE LOAD

TIGHT WAD

TYPE CRIME

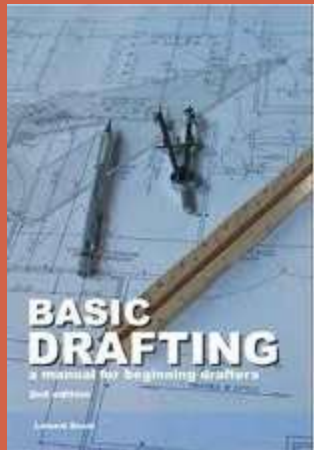
HORIZONTAL & VERTICAL SCALING

The proportions of the letters have been digitally distorted in order to create wider or narrower letters.



LIGHTFACE LETTERS

Letters with very thin
stems.

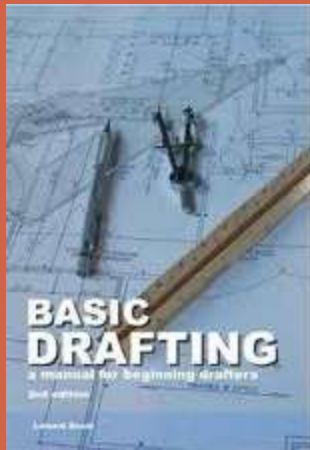


informative
informative



BOLDFACE LETTERS

Letters with heavy stems.

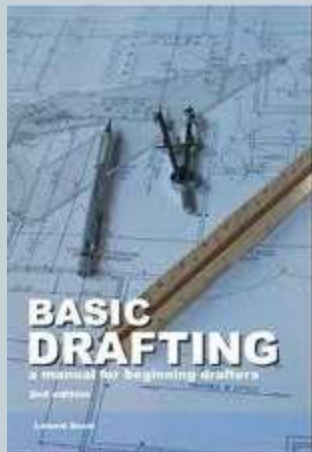


informative
informative

GUIDE LINES



- Extremely light horizontal lines are necessary to regulate the height of letters. In addition, light vertical or inclined lines are needed to keep the letters uniformly vertical or inclined.



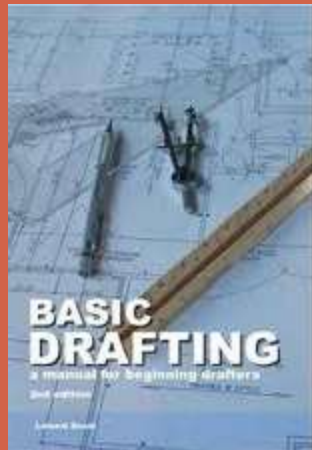
PARTS OF GUIDE LINES

CAP LINE

WAIST LINE

BASE LINE

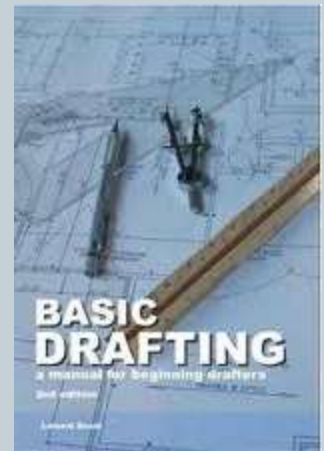
DROP LINE



TYPES OF DRAWING

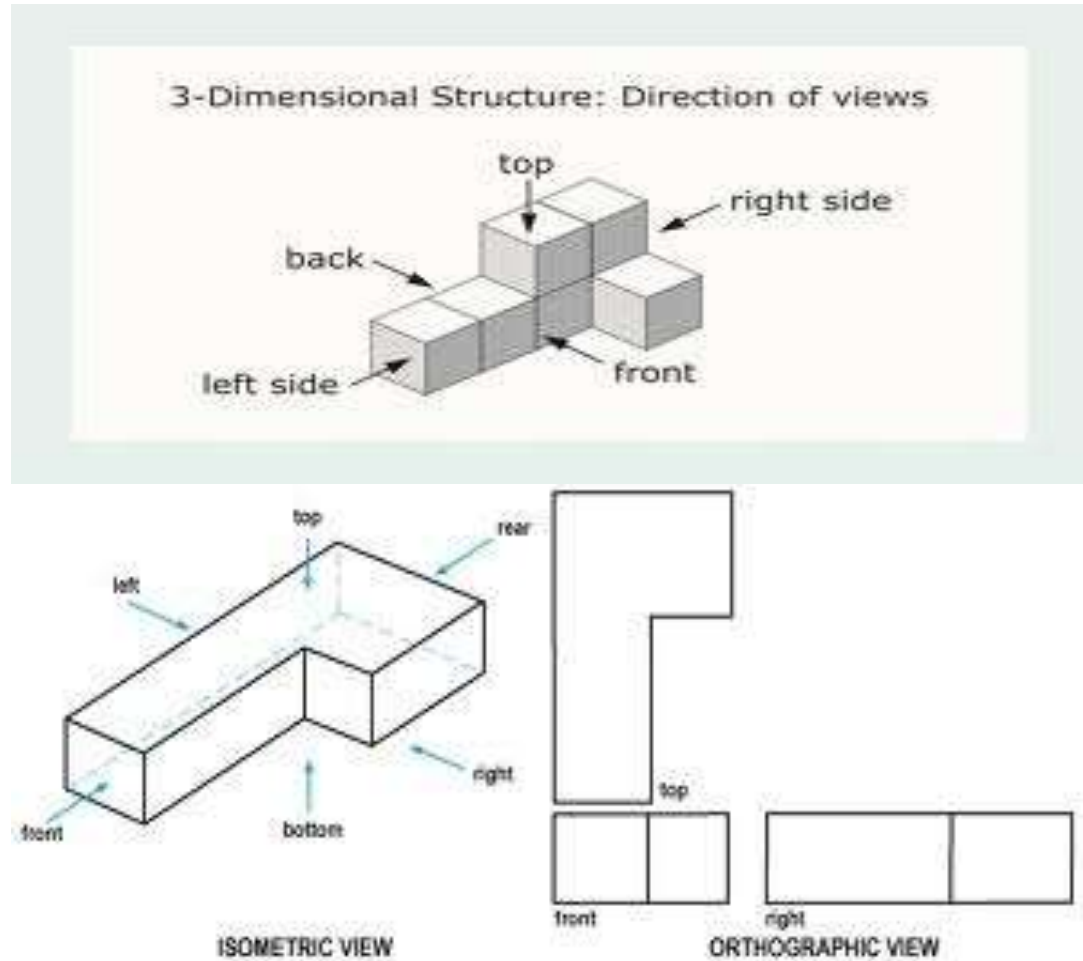


- ORTHOGRAPHIC DRAWING
- ISOMETRIC DRAWING
- PICTORIAL DRAWING
 - AXONOMETRIC
 - ✦ ISOMETRIC
 - ✦ DIMETRIC
 - ✦ TRIMETRIC
 - OBLIQUE
 - ✦ CAVALIER
 - ✦ CABINET
 - PERSPECTIVE
 - ✦ PARALLEL
 - ✦ ANGULAR



ORTHOGRAPHIC DRAWING

A drawing that shows the principal views of an object. There are six views in an object; top view, front view, bottom view, back view, left side view and right side view. In orthographic drawing only three views are necessary to project; these are the top view, front view and side view.

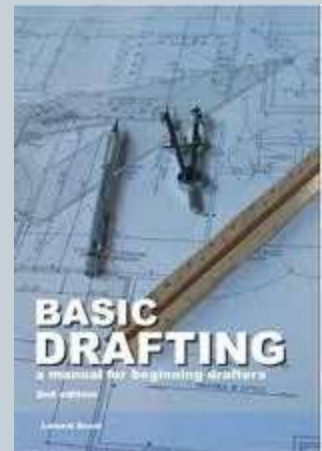
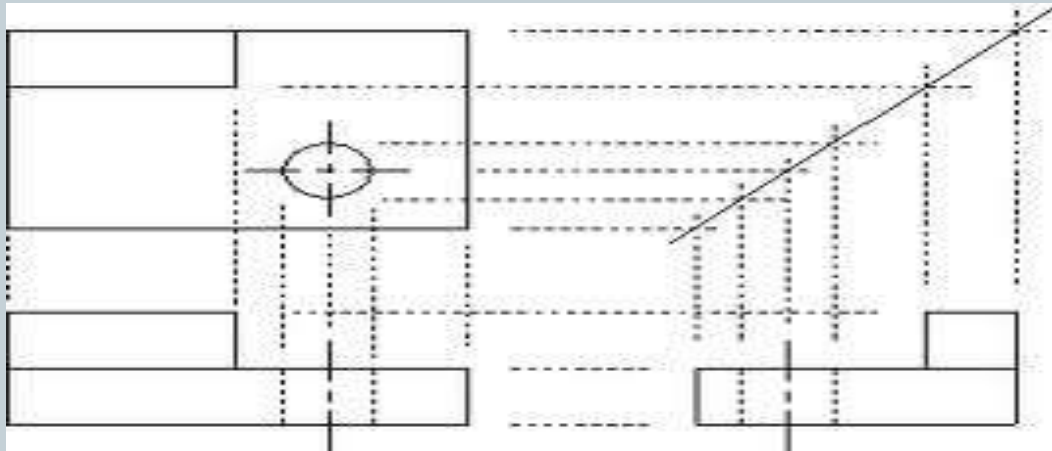


To make an orthographic drawing, there should be a given pictorial drawing.

PRINCIPLES OF ORTHOGRAPHIC PROJECTION



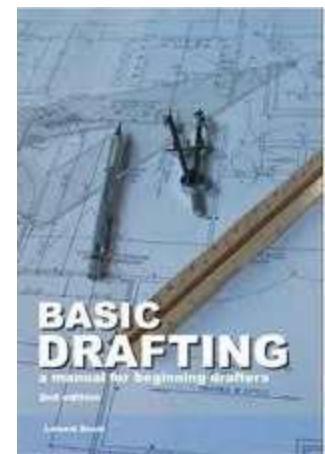
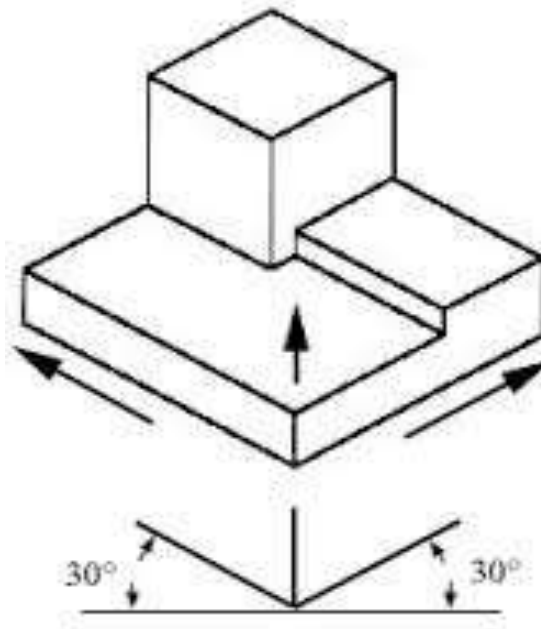
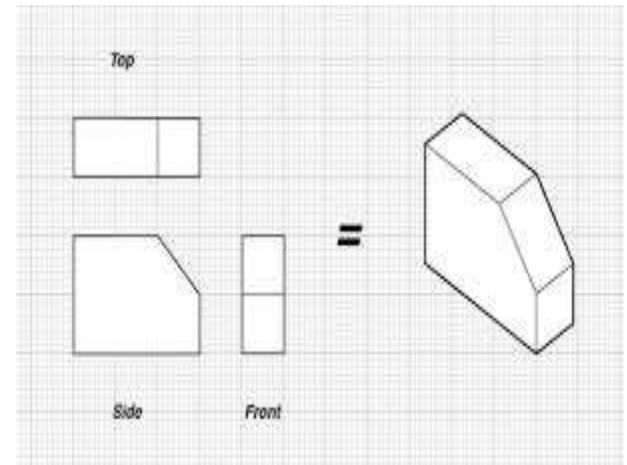
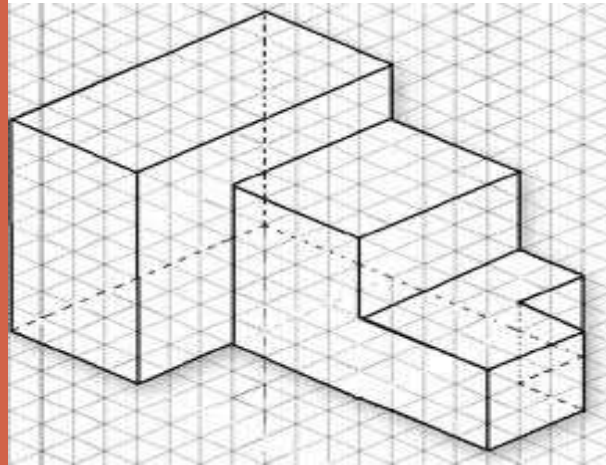
- The top view is directly drawn above the front view.
- The side view is drawn horizontally in line with the front view.
- A line parallel to a plane of projection will show its true length on that plane.



ISOMTERIC DRAWING

Is a drawing wherein the object is represented by three sides seen at one time. It is made with 30x60 degrees triangle. It is also a three dimensional drawing because height, width and depth are all drawn in a single plane.

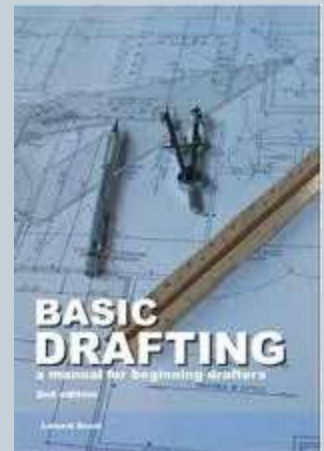
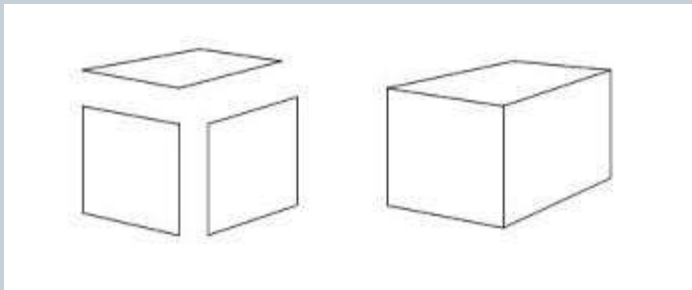
Box method is the starting point for all isometric drawings.



PRINCIPAL DIMENSIONS



- **WIDTH-** is the measurement of an object taken from side to side of an object.
- **HEIGHT-** is the measurement taken from top to bottom of an object.
- **DEPTH-** is the measurement from front to rear of an object.

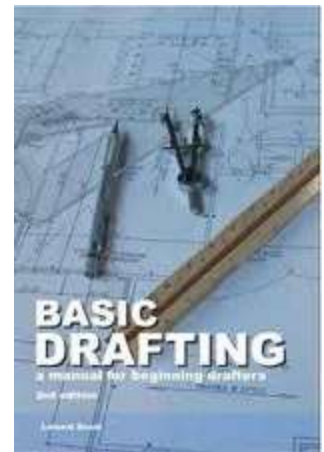
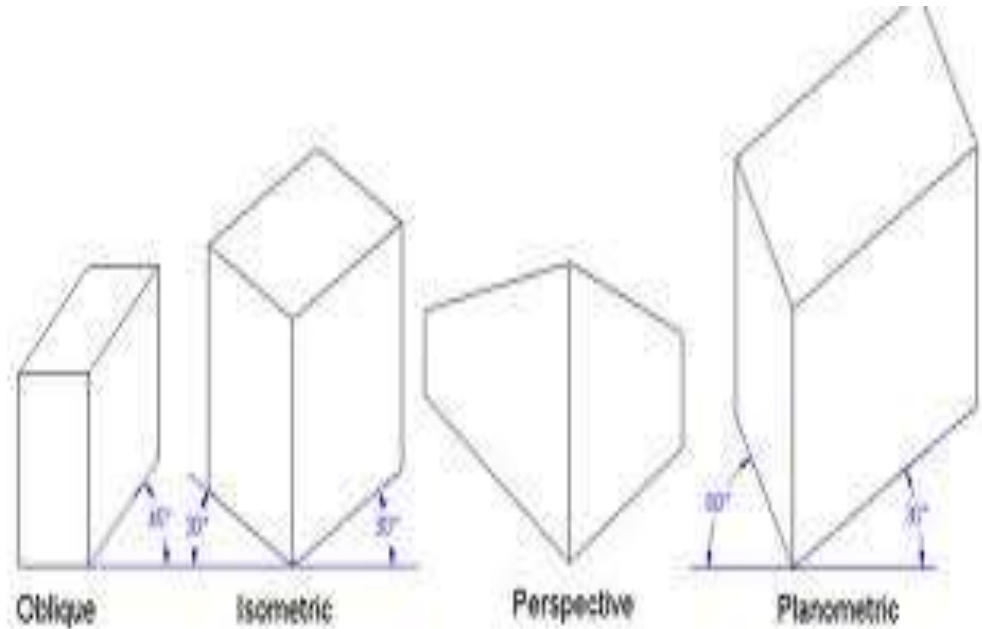


PICTORIAL DRAWING

Are drawings that show objects as they would appear in a reality. It is also a drawing that shows more than one side of the object.

TYPES:

- AXONOMETRIC
- OBLIQUE
- PERSPECTIVE



AXONOMETRIC DRAWING

Has the object turned so that the three faces are viewed simultaneously. The axonometric positions are:

Isometric

Dimetric

Trimetric

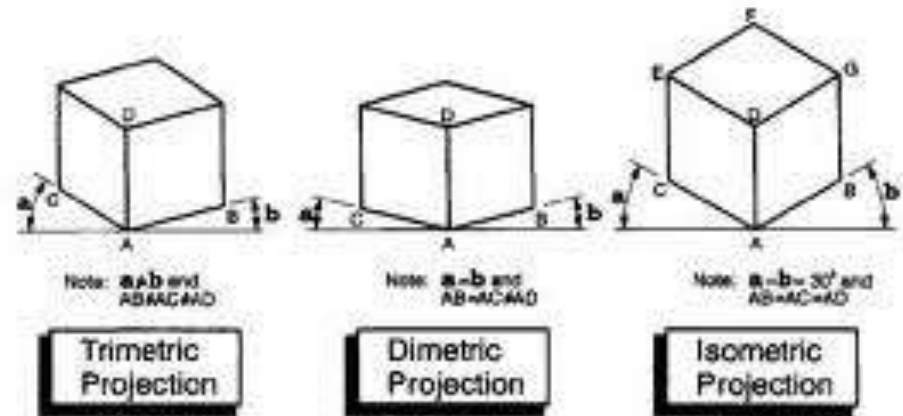
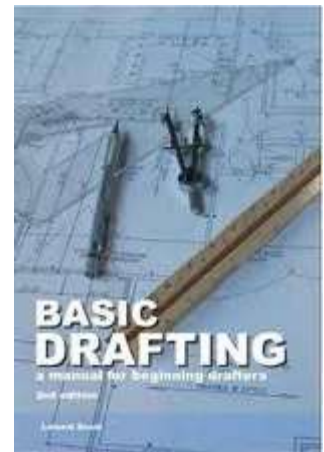


Figure 2.4 *The three types of axonometric projections*





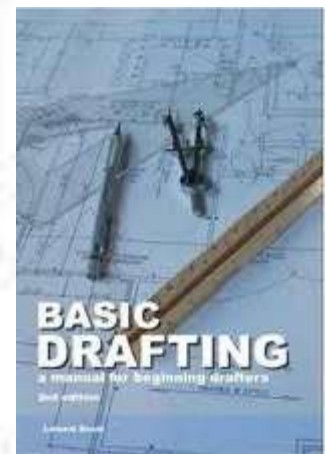
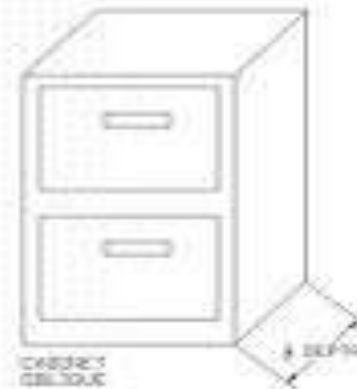
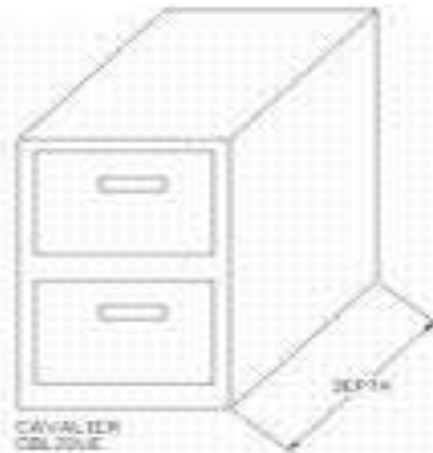
OBLIQUE DRAWING

Have one plane parallel to the picture plane which is the same size as the orthographic view.

TYPES:

- CAVALIER
- CABINET

OBLIQUE DRAWING



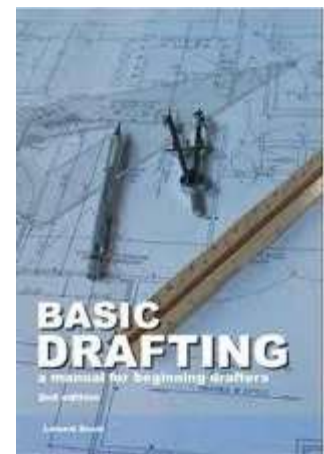
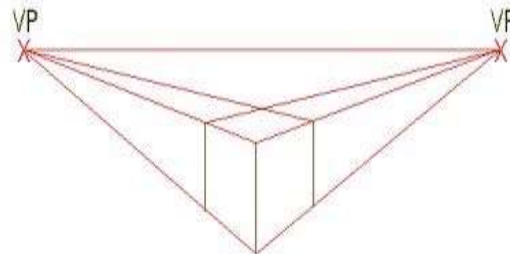
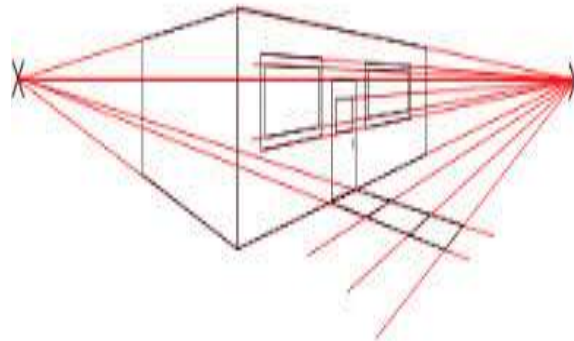
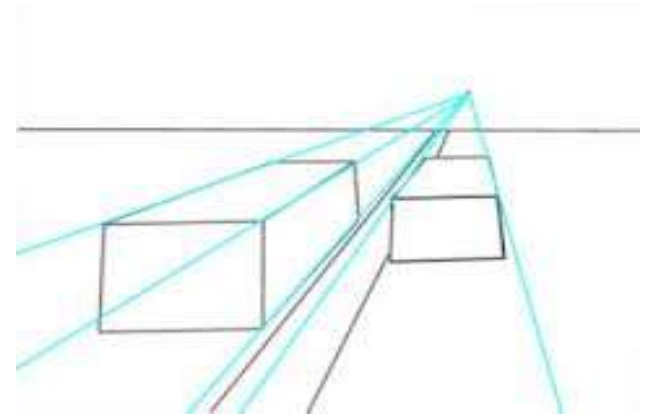
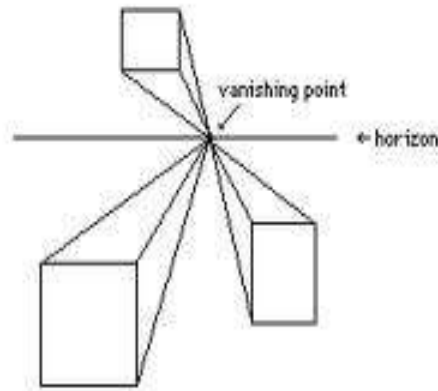
PERSPECTIVE DRAWING

Is the pictorial view as seen by the observer's eyes.

TYPES:

-PARALLEL OR ONE POINT PERSPECTIVE

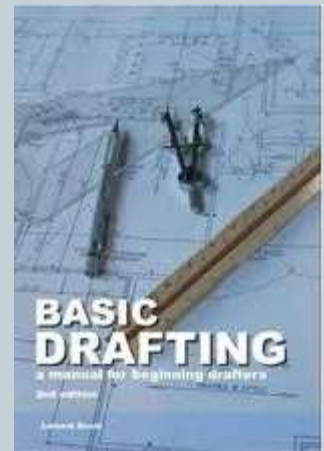
-ANGULAR OR TWO POINT PERSPECTIVE



ADDITIONAL INFORMATION



- OBJECT is a figure being projected and PROJECTION is a way of expressing and showing the object.



DIMENSIONING A DRAWING



- **DIMENSION**- Tells the accurate shape and size of an object.
- **DIMENSION LINE**- A line that contrasts well with the outline of the object.
- **EXTENSION LINE**- A fine line used to show clearly the dimension limits.
- **CENTER LINE**- are light broken line which are sometimes used as an extension line.
- **LEADER LINE**- are straight line leading from a dimension value.
- **ARROWHEADS**- indicate the direction in which the value applies.
- **NUMERICAL VALUE**- gives the actual distance of the objects.
- **NOTES**- provide a means of explanatory information along with a size.